

Piotr Brozek

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EDUCATION

University of Illinois at Chicago

Expected May 2026

Bachelor of Science in Computer Science; GPA: 3.8

Chicago, IL

- **Relevant Coursework:** Machine Learning, Algorithms and Data Structures, Software Design
- **Extracurriculars:** Association for Computing Machinery, CodePath Technical Interview Prep

EXPERIENCE

Research Assistant

Aug. 2024 – May 2025

Electronic Visualization Laboratory

Chicago, IL

- Collaborated with the software engineering team to enhance SAGE3, a collaborative whiteboard used by 5,000+ users across 800 institutions, using TypeScript, React, LangChain, and Docker
- Implemented LLM features including DeepPlot visualizations, speaker diarization, and Llama 3 summarization to improve transcription accuracy and reduce manual summarization by 50%
- Developed AI-powered export tools with Python and React to generate summaries as CSVs, PDFs, images, and sticky notes

Full-Stack Developer Intern

Oct. 2024 – Feb. 2025

Ingenio Healthcare

Lombard, IL

- Developed a physician-finder tool with React, Redux, Node.js, PostgreSQL, MapGL, and Material UI to help users locate providers and book appointments
- Built an AI-powered insurance authorization system using the OpenAI API and PostgreSQL, reducing processing time by 30%
- Contributed to frontend development of new features using Material UI, enhancing user experience and interface responsiveness across the platform

Full-Stack Developer Intern

June 2024 – Aug. 2024

Electronic Visualization Laboratory

Chicago, IL

- Improved the speed and accuracy of an AI-powered stock dashboard by 25% by implementing custom web scraping functions and filtering data from over 100 sources per request.
- Designed and developed a live stock dashboard with real-time data analysis using React, Node.js, Express, and AI integration for predictive insights.
- Created innovative web applications including a movie website using the Fetch API and an AI-powered chatbot to enhance user engagement and functionality.

PROJECTS

MRI Brain Tumor Detection and Classification | *Python, TensorFlow, Keras, NumPy*

- Developed a machine learning model in Python to classify brain tumors in MRI scans using over 10,000+ images, combining ResNet-18 for deep feature extraction with AdaBoost to boost accuracy on small clinical datasets.
- Preprocessed data with Pandas, applied transfer learning, and trained models on GPU with cross-validation, hyperparameter tuning, and performance visualized via Matplotlib.
- Demonstrated that the hybrid approach significantly improved classification accuracy in limited-data scenarios compared to standalone CNNs.

AI-Powered Stock Dashboard | *React, TypeScript, Express, Vite, TailwindCSS, Chakra UI*

- Implemented web scraping for 50–100 news articles to generate stock predictions via Llama 3 using Axios.
- Integrated NewsAPI and Finnhub API for real-time stock and news data.
- Built full-stack dashboard with Node.js backend and React frontend.

TECHNICAL SKILLS

Languages & Databases: JavaScript, TypeScript, Python, Java, F#, C/C++, Go, SQL, PostgreSQL, MongoDB

Frameworks & Libraries: React, Node.js, Express, Next.js, LangChain, Docker, PyTorch, TensorFlow

Tools & Platforms: Jupyter Notebook, Linux/Unix, VS Code, IntelliJ, Eclipse, GDB, GoogleTest, VirtualBox